

Project Initialization and Planning Phase

Date	29 September 2026
Team ID	SWTID-2026-1156
Project Title	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) Report

EduGenie is an AI-powered learning assistant designed to make studying easier, faster, and more personalized. It uses the Google Gemini API to provide simple explanations, answer academic questions, summarize topics, generate study support, and help students revise according to their learning needs.

Project Overview

Objective	To develop an intelligent learning assistant that provides quick, clear, and personalized academic support using Google Gemini.
Scope	The project covers question answering, concept explanations, topic summaries, study assistance, and personalized learning support through a user-friendly web application.

Problem Statement

Description	Students often face difficulty understanding complex concepts, finding simple explanations, revising large amounts of content, and getting immediate help when teachers or tutors are unavailable.
Impact	These difficulties can increase study time and confusion. A quick and personalized learning assistant can improve understanding, revision, accessibility, and overall learning efficiency.

Proposed Solution

Approach	Integrate Google Gemini with a web-based learning application so students can ask questions and receive context-aware, easy-to-understand responses.
Key Features	• AI-powered question answering and concept explanation. • Topic and lesson summarization. • Personalized study assistance and revision support. • Interactive chat-based learning interface. • Fast responses through the Gemini API.

Project Resources

Resource Type	Description	Specification / Allocation
Hardware		
Computing Resources	Computer/laptop for development, testing, and deployment preparation.	Minimum 4 GB RAM; internet connection
Memory	RAM required for development tools and application testing.	8 GB recommended
Storage	Storage for source code, project files, datasets, and logs.	At least 20 GB free space
Software		
Frameworks	Web application and backend development.	HTML, CSS, JavaScript, Python Flask
Libraries / APIs	AI integration and supporting application libraries.	Google Gemini API, Python SDK/libraries
Development Environment	IDE and source-code management tools.	VS Code / PyCharm, Git, GitHub
Data		
Data	Learning prompts, user queries, topic information, and optional user learning history.	Structured application data; no large training dataset required for Gemini API usage

Expected Outcome: EduGenie will provide students with an accessible AI learning companion that reduces the time needed to find explanations and supports clearer, more personalized study.